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M2

Started on	Saturday, 12 September 2026, 9:45 PM
State	Finished
Completed on	Saturday, 12 September 2026, 9:46 PM
Time taken	37 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findPair(int arr[], int left, int right, int x, int *a, int *b)
4  {
5      if (left >= right)
6          return 0;
7
8      int sum = arr[left] + arr[right];
9
10     if (sum == x)
11     {
12         *a = arr[left];
13         *b = arr[right];
14         return 1;
15     }
16
17     if (sum < x)
18         return findPair(arr, left + 1, right, x, a, b);
19
20     return findPair(arr, left, right - 1, x, a, b);
21 }
22
23 int main()
24 {
25     int n, x;
26     scanf("%d", &n);
27
28     int arr[n];
29
30     for (int i = 0; i < n; i++)
31         scanf("%d", &arr[i]);
32
33     scanf("%d", &x);
34
35     int a, b;
36
37     if (findPair(arr, 0, n - 1, x, &a, &b))
38     {
39         printf("%d\n", a);
40         printf("%d\n", b);
41     }
42     else
43     {
44         printf("No");
45     }
46
47     return 0;
48 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course